EE256: ROOFTOP SOLAR PLANT DESIGN AND ANALYSIS

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Task 1: Characterizing the Building and Rooftop Solar Plant Design

Objective: Design a rooftop solar PV system for a hostel building based on site-specific data and engineering principles.

a) Site Assessment:

- Use Google Maps, Google Earth, or equivalent software to identify the hostel building assigned to you.
- Determine the approximately south-facing roof area (in square meters) suitable for solar panel installation. Account for obstructions and shading from nearby structures or trees.
- Provide a screenshot or diagram of the roof with annotations indicating usable area and any excluded sections.

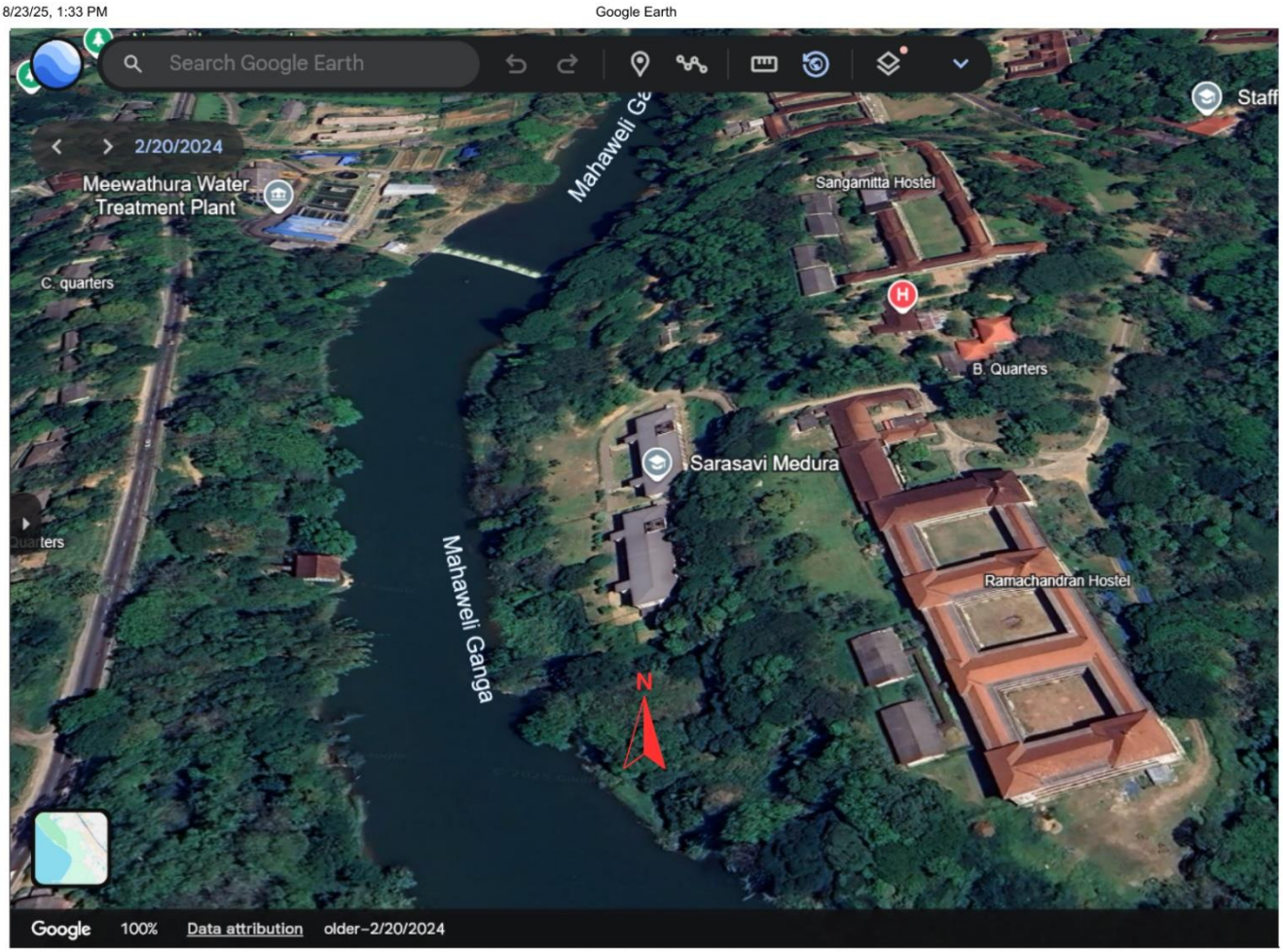


Figure 1 : Bird's eye view of Sarasavi Madura Hostel



Figure 2 : Sarasavi Madura Hostel

In our hostel buildings, south facing roof areas are very small. Also for east direction, more sun light is covered from the mountain. There are some areas in the building roof which obstruct for solar system. (water tank area, shading from nearby trees)

So, we are getting west side roof area for install the solar system.

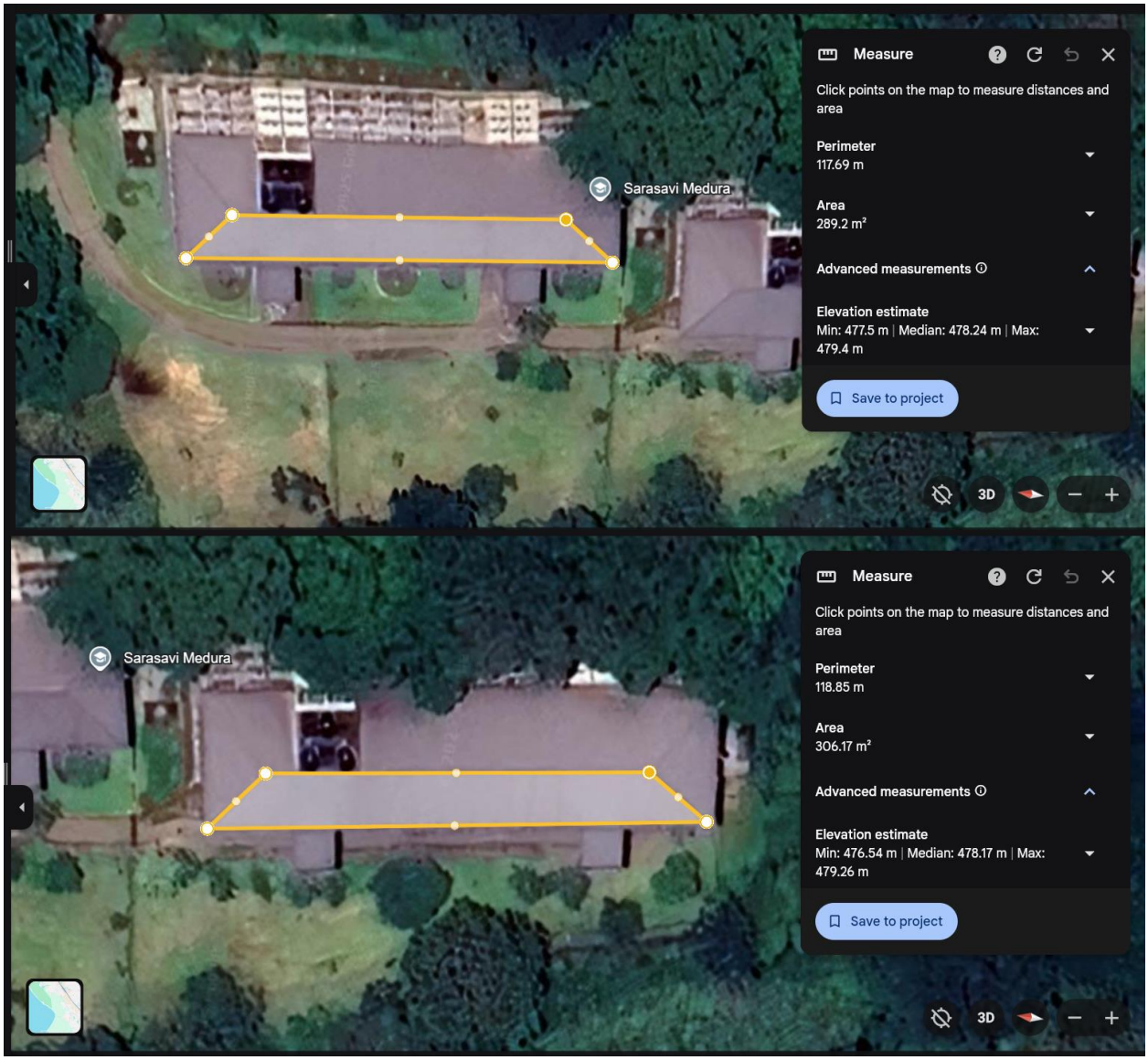


Figure 3: Measurements of Installing roof area

Roof area for building 1 = 289.2 m^2
Roof area for building 2 = 306.2 m^2

Total roof area = 595.4 m^2

b) Solar Plant Design:

- Using design techniques from the industrial lecture (e.g., panel sizing, spacing to avoid shading, and system capacity optimization), propose a suitable PV system.
- Select appropriate solar panels (specify type, efficiency, and wattage) and inverters (specify type and capacity) based on industry standards.
- Calculate the total system capacity (in kW) and the number of panels required, considering the usable roof area and panel dimensions.

With conservative allowances for edges, walkways, vents and obstructions ≈15% loss,

So, the effective PV area is:

Effective PV area = 595.4*0.85 = 506.1 m^2

Packing with access & setbacks already accounted for above. So, target module area is less than or equal to 506.1 m^2.

In general, 2333 x 1134 x 30mm panel is available from Hayleys

One module area = 2.6 m^2
Total applicable module count = 506.1 / 2.6 = 194

So, we use 192 modules for a better design.

Distribute approximately by area:

- Building 1 → 96 modules
- Building 2 → 96 modules
  (Adjusted on site to avoid skylights and obstructions)

Capacity = 192 *545 W = 104.6 kW (DC)

Module choice :

- Module class: 545 W mono N-type Topcon, half-cut, 144 cells
- Efficiency: 21%
- Dimensions (Area): 2333 × 1134 mm = ~2.6 m^2 per module

Inverters :

- Type: single-phase, grid-tied string inverters, outdoor-rated (IP65+), each with multiple MPPTs and arc-fault protection.
- Two 50kW inverters
- Capacity = 2 × 50 kW = 100 kW (AC)

String design for ~545 W modules :

Typical module: Voc ≈ 49–50 V, Vmp ≈ 41–42 V.

- Keep ≤ 1000–1100 Vdc limit → 16 modules/string
- Strings required: 192 modules / 16 = 12 strings.
- With two 50 kW inverters (each with ≥6 MPPTs), use 1 string per MPPT for best MPP tracking:
  - Inverter 1: 6 MPPTs × 1 string each = 6 strings
  - Inverter 2: 6 MPPTs × 1 string each = 6 strings

Total system capacity = 192 * 0.545 kW = 104.64 kW (DC)

c) Electrical Layout Diagram:

- Draw a single-line electrical diagram showing:
  - Arrangement of PV modules (series/parallel connections).
  - Inverter placement and connection to the grid.
  - Metering and safety devices (e.g., circuit breakers, disconnect switches)

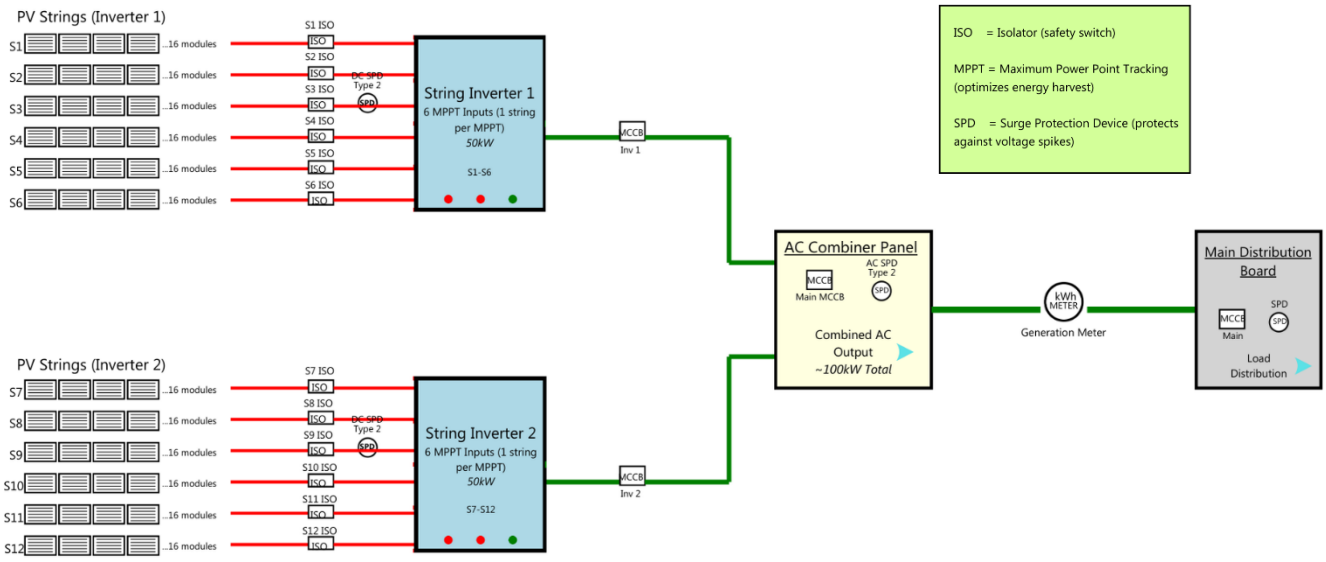


Figure 4 : Single Line Diagram

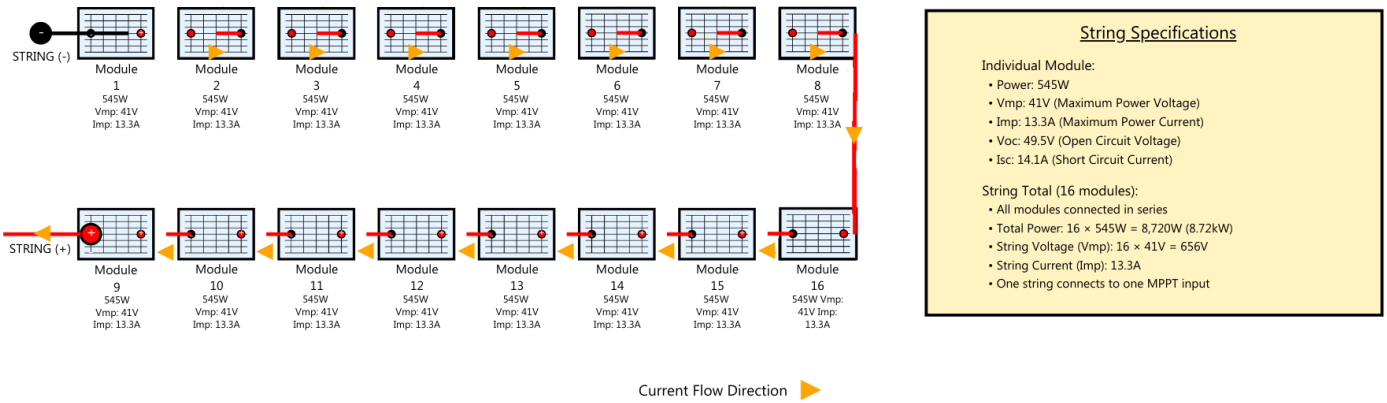


Figure 5 : 16-Module PV String - Series Connection

d) Daily Energy Demand Estimation:

- Estimate the hostel's daily electrical demand (in kWh) for weekdays and weekends.
- Divide the day into logical time periods (e.g., morning: 6 AM–12 PM, afternoon: 12 PM–6 PM, evening: 6 PM–12 AM, night: 12 AM–6 AM).
- Provide a table summarizing demand for each time period, considering typical hostel activities.
- Justify assumptions based on typical hostel occupancy and equipment usage.

There are 96x2 rooms and 3 students per Room = 576 students

Loads (diversified): 2×10 W LED lights, 2×7 W wall lamps, charging/laptop
≈ 75 W per student

Daily max power demand for a room = 20 + 14 + 150
= 184 W

Daily max demand for all rooms = 184*96*2
= 35.3 kW

Assuming 20% students staying at the hostel in normal weeks like without exams
For weekend days = 35.3*0.2
≈ 7 kW

Other demands are like floor lightings and canteen (only for lights), etc..

- Assuming there are seven 12W LEDs for each floor includig wash rooms, there are total 8 floors

So, power for daily floor lights = 12*7*8
= 672 W

- So, with canteen, etc; let's assume max demand for other loads is
= 1 kW

Table 1 : Daily Energy for a Week Day

Time Period	Room Load (kW)	Other Load (kW)	Total Avg Load (kW)	Energy (kWh)
6 AM–12 PM	5	0.3	5.3	31.8
12 PM–6 PM	7	0.5	7.5	45.0
6 PM–12 AM	35.3	1	36.3	217.8
12 AM–6 AM	10	1	11	66.0
Total	57.3	2.8	60.1	360.6

Table 2 : Daily Energy for a Weekend Day

Time Period	Room Load (kW)	Other Load (kW)	Total Avg Load (kW)	Energy (kWh)
6 AM–12 PM	1	0.3	1.3	7.8
12 PM–6 PM	1.4	0.5	1.9	11.4
6 PM–12 AM	7	1	8	48.0
12 AM–6 AM	2	1	3	18.0
Total	11.4	2.8	14.2	85.2

Task 2: Solar Resource Data Collection

Objective: Gather and analyse solar resource data for the hostel's location using PVGIS.

a) Location Details:
- Provide the hostel's address and its precise latitude and longitude (use Google Maps or PVGIS for accuracy).

b) Data Collection:
- Using PVGIS, complete the following table for the hostel's location:

Table 3 : Details of the Hostel

Home address	Sarasavi Medura Hall, University of Peradeniya, Peradeniya, 20400.	
Parameter	Units	Home location
Latitude	^{0}	7.246476
Longitude	^{0}	80.596137
Annual global insolation (irradiation) on an optimally inclined plane	kWh/m^2.year	1867
Tilt angle for maximum annual insolation	^{0}	6
Daily insolation (irradiation) in December at an optimum tilt angle	kWh/m^2.day	3.98
Daily insolation (irradiation) in June at an optimum tilt angle	kWh/m^2.day	5.31
Ratio of diffuse/global insolation throughout the year	%	34.083

Daily insolation (irradiation) in December at an optimum tilt angle = 123.41/31
= 3.98 kWh/m^2.day

Daily insolation (irradiation) in June at an optimum tilt angle = 159.16/30
= 5.31 kWh/m^2.day

Ratio of diffuse/global insolation throughout the year,

=(0.34 + 0.32 + 0.31 + 0.25 + 0.33 + 0.31 + 0.35 + 0.26 + 0.41 + 0.39 + 0.42 + 0.40) / 12
= 0.34083

NOTE: The final year for which data is available is 2023. Therefore, this report is based on data obtained during the year 2023

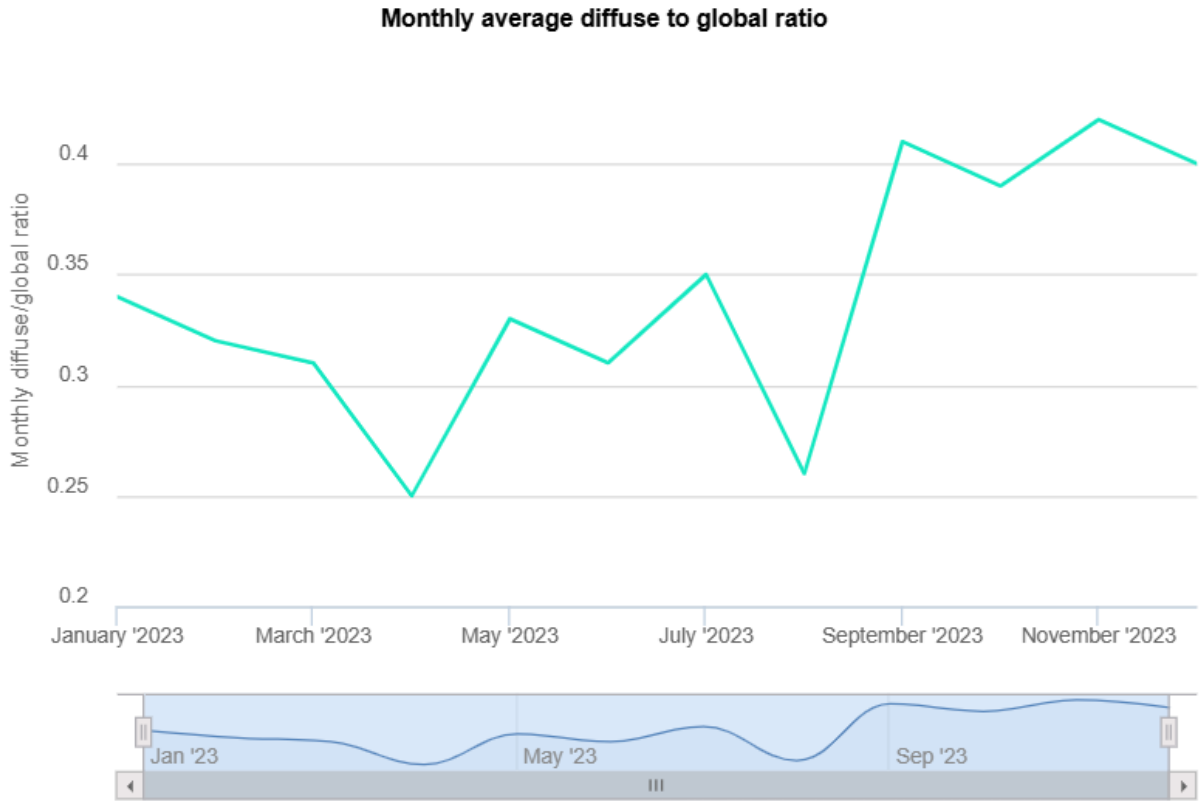


Figure 6 : Monthly Average Diffuse to Global Ratio

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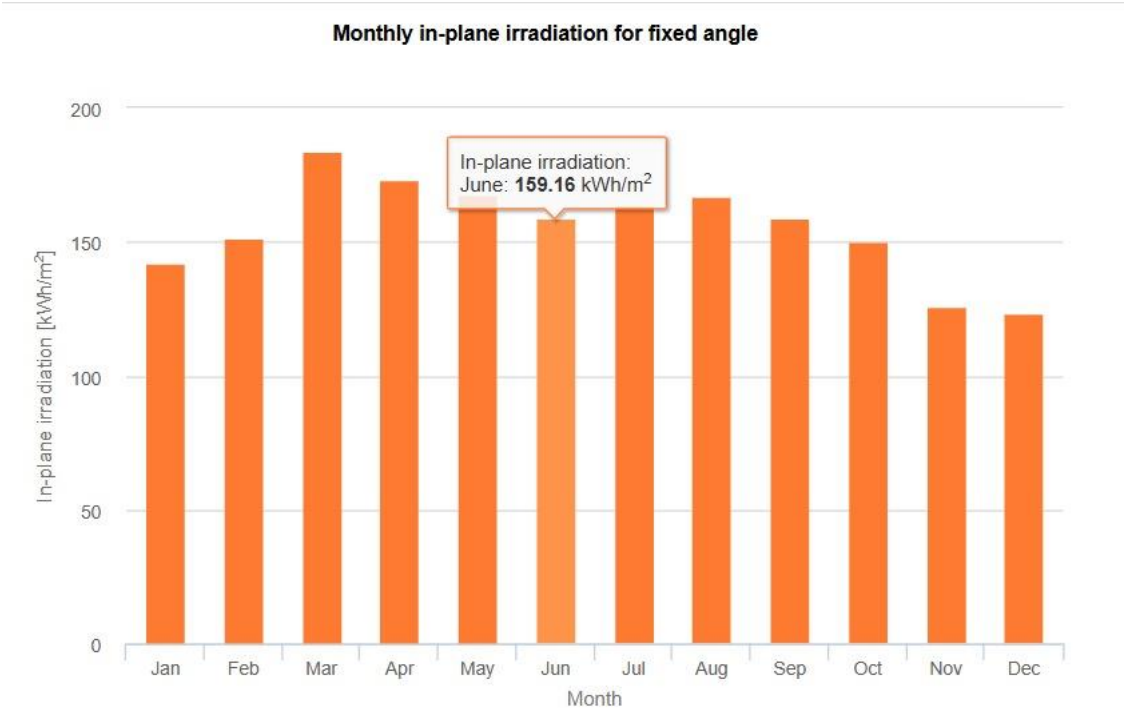


Figure 7 : Monthly In-Plane Irradiation for Fixed Angle (June)

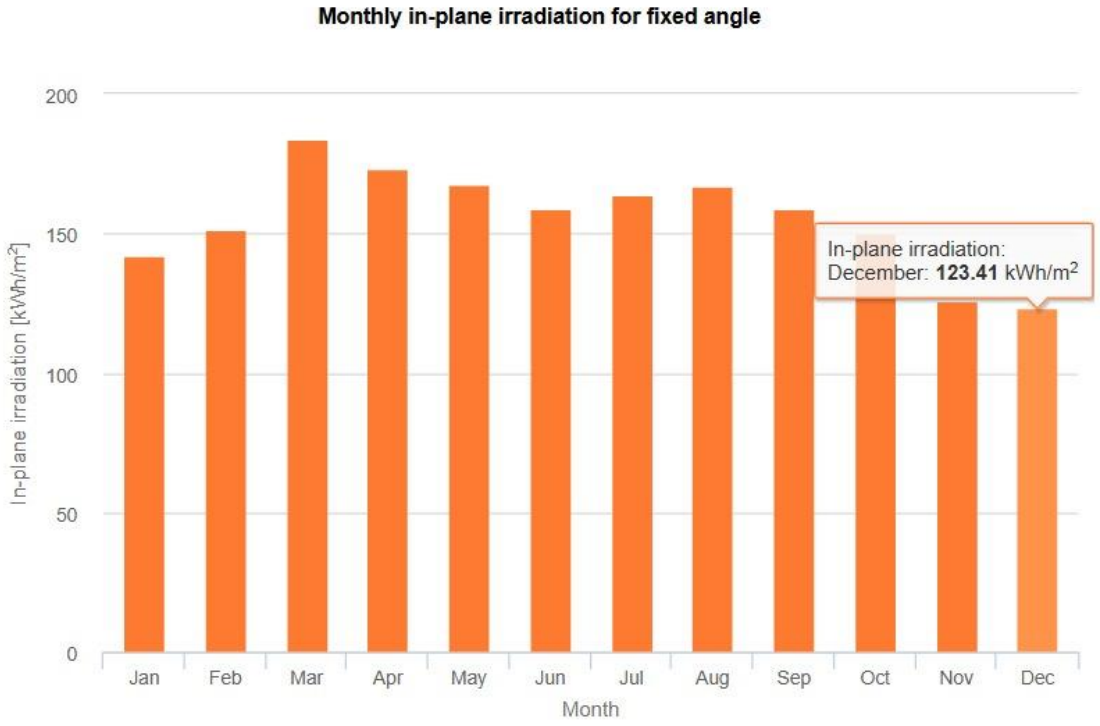


Figure 8 : Monthly In-Plane Irradiation for Fixed Angle (December)

Task 3: Solar Resource Visualization

Objective: Visualize and interpret solar resource data to understand seasonal and daily variations.

a) Data Collection:
- Use PVGIS to extract monthly and daily solar irradiation data for the hostel's location at the optimum tilt angle.

b) Graphical Analysis:
- Create the following graphs using software like Excel, Python (Matplotlib), or PVGIS's built-in tools:
  - Graph 1: Monthly global insolation (kWh/m²) at the optimum tilt angle for all 12 months.
  - Graph 2: Average daily irradiance (W/m²) over 24 hours for March, August, and December at the optimum tilt angle.
- Ensure graphs are clearly labelled (title, axes, units, legend) and include a brief description of trends observed (e.g., seasonal variations, peak irradiance times).

Provided inputs:	
Location [Lat/Lon]:	7.246,80.596
Horizon:	Calculated
Database used:	PVGIS-ERA5
PV technology:	Crystalline silicon
PV installed [kWp]:	104.64
System loss [%]:	14

Simulation outputs:	
Slope angle [°]:	6 (opt)
Azimuth angle [°]:	-90
Yearly PV energy production [kWh]:	144300.41
Yearly in-plane irradiation [kWh/m^2]:	1867
Year-to-year variability [kWh]:	2956.89
Changes in output due to:	
Angle of incidence [%]:	-2.47
Spectral effects [%]:	1.02
Temperature and low irradiance [%]:	-12.82
Total loss [%]:	-26.14

Figure 9 : Summary of Data obtained from PVGIS

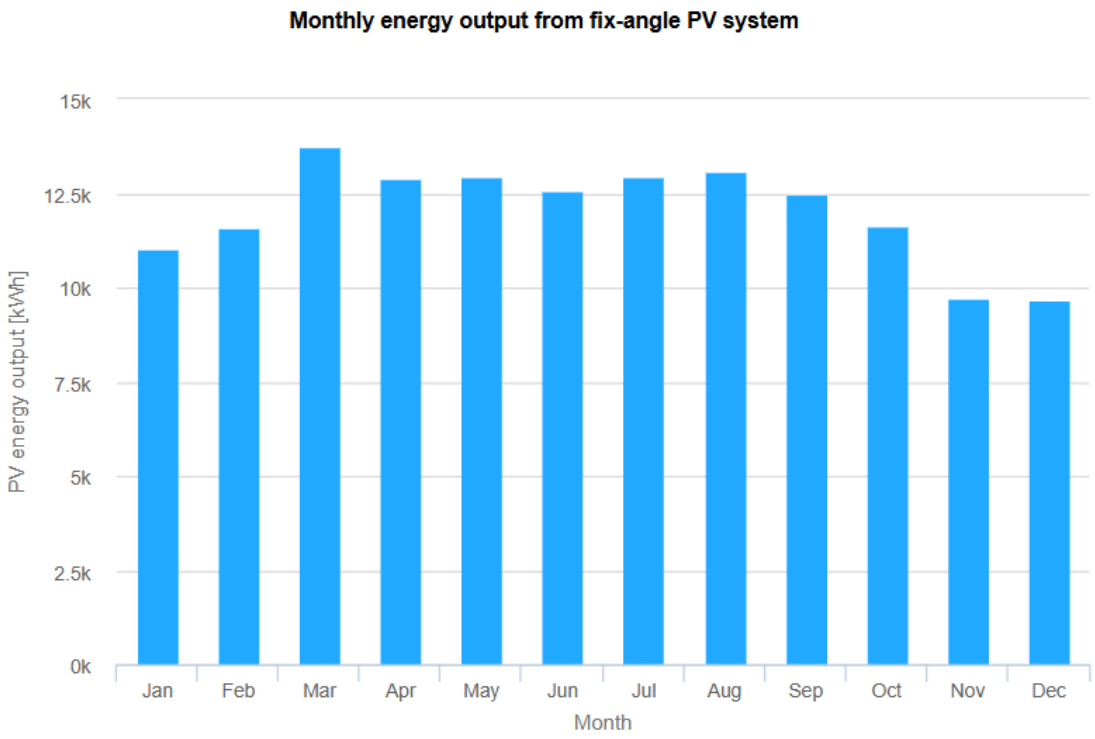


Figure 10: Monthly Energy Output from Fix-Angle PV System

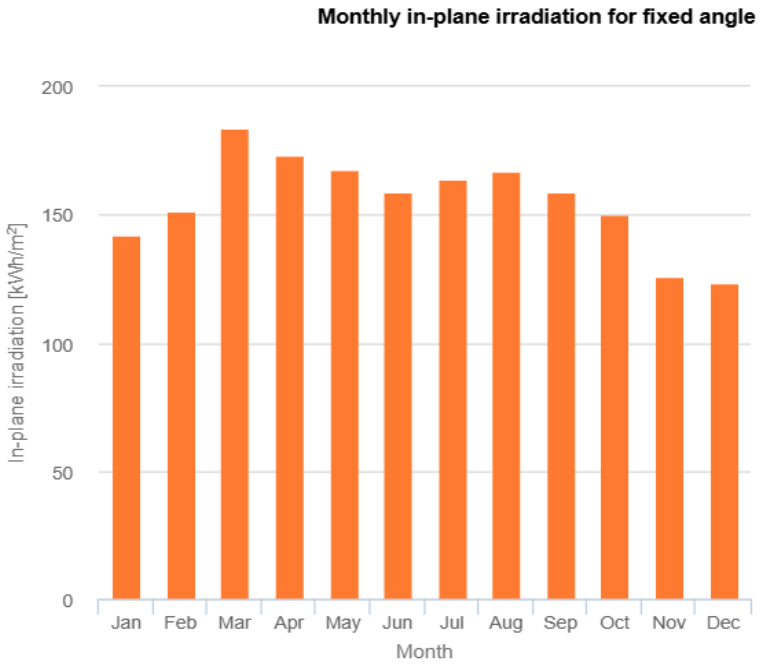


Figure 11 : Monthly In-Plane Irradiation for Fixed Angle

The graphs for March, August, and December are displayed below. They all show the characteristic bell-shaped curve of daily solar irradiance but with significant variations in intensity and duration.

For all 3 graphs, there is a clear and strong seasonal variation in both the peak irradiance and the daily energy total (the area under the "Global" curve).

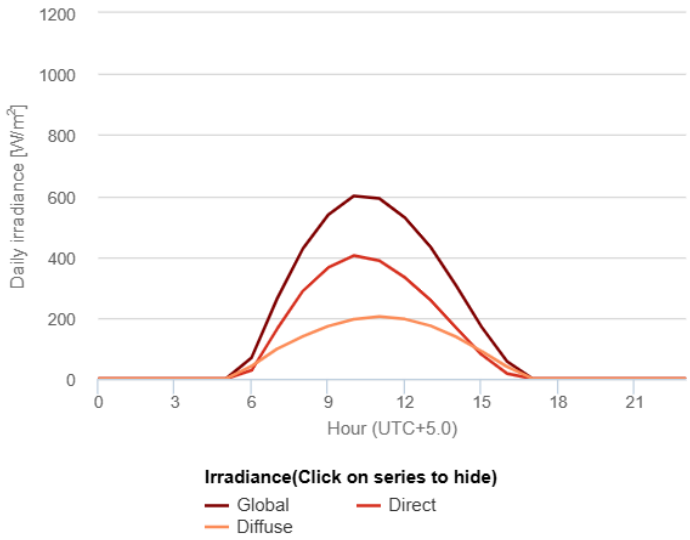


Figure 12 : Daily Average Irradiance for December

December has the lowest overall irradiance values and the shortest "daylight" window (from ~06:30 to 16:30).

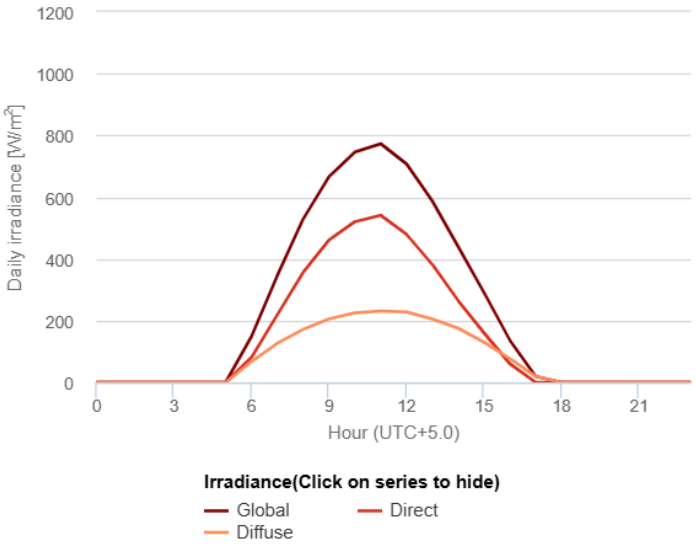


Figure 13 : Daily Average Irradiance for August

August shows strong, sustained irradiance throughout the middle of the day, with a very broad peak.

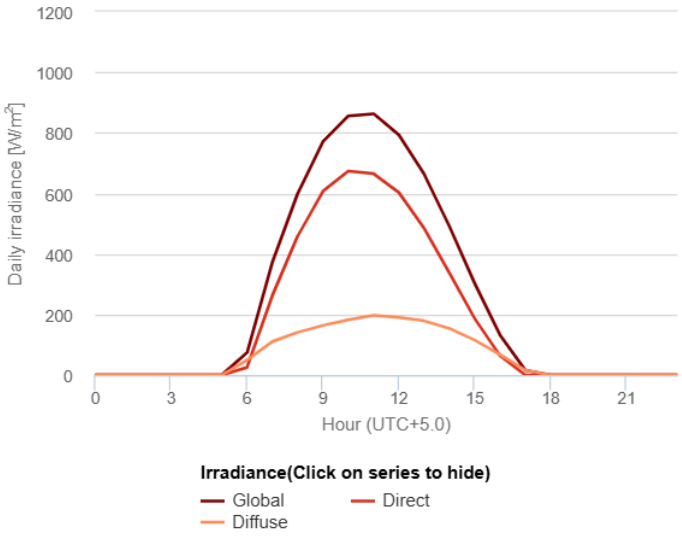


Figure 14 : Daily Average Irradiance for March

March exhibits the highest peak irradiance values (nearly 860 W/m² at 11:30).

Peak Irradiance Times: For all three months, the peak of global irradiance occurs around solar noon, between 11:30 and 12:30 (UTC+5). This is consistent throughout the year.

Direct vs. Diffuse Composition:

March has a high proportion of Direct irradiance, especially around noon, indicating clear sky conditions.

August also has a strong Direct component, but the Diffuse component is more significant than in March, suggesting the possibility of more atmospheric haze or thin clouds.

December has a significantly higher Diffuse-to-Direct ratio. The diffuse component is a much larger fraction of the global total, indicating cloudier conditions or a lower sun angle increasing atmospheric scattering.

c) Interpretation:
   - Discuss how seasonal changes in insolation affect PV system design and performance.
   - Highlight differences in irradiance patterns between March, August, and December, and their implications for energy production.

How Seasonal Changes in Insolation Affect PV System Design and Performance:

System Sizing and Energy Yields the primary impact is on energy production estimates. A PV system will produce significantly more energy in March and August than in December.

The sun's path is higher in the sky in March and August and lower in December. The fixed tilt of 6° in this data is very shallow due to the proximity of the location to the equator and is optimized for annual yield.

Also, the panel efficiency decreases with temperature. Although irradiance is highest in March, the accompanying higher ambient temperatures (implied by the data in previous files) can slightly reduce conversion efficiency compared to the cooler days of December. This temperature effect must be factored into performance predictions.

The seasonal mismatch between production (high in dry seasons, lower in wet seasons) and potential demand (e.g., higher for cooling in August) is a critical design factor. It often necessitates a connection to the electrical grid or a very large battery storage system to shift energy from high-production to high-consumption periods. However due to the scale and purpose of the plant in this instance there will not be a battery storage system.

Table 4 : Differences in irradiance patterns between March, August, and December and their implications for energy production

Feature	March	August	December	Implication for Energy Production
Peak Global Irradiance	Highest (~863 W/m²)	High (~773 W/m²)	Lowest (~600 W/m²)	March has the highest potential for peak power output. December has the lowest instantaneous power generation.
Curve Shape	Sharp, symmetric peak	Broad, sustained peak	Narrow, lower peak	August has more hours of high irradiance, leading to a high daily energy total. December's short duration severely limits daily production.
Direct Component	Strongest & Dominant	Strong	Weakest	March is ideal for technologies that require direct sunlight (e.g., Concentrated Solar Power - CSP). All months are good for standard PV, but March and August are superior.
Diffuse Fraction	Lowest	Moderate	Highest	December's output is more dependent on diffuse light. Systems will perform less efficiently than under the clear skies of March, as diffuse light is harder to concentrate and utilize optimally.
Daylight Hours	Long	Long	Short	This is the most important factor. December has far fewer sun hours, which is the main reason its total daily energy production (kWh) is roughly 40-50% lower than in March or August.

Therefore, a PV system at this location will experience high, reliable output from March through August with March offering the highest peak power and August offering sustained high output throughout the day. December represents a period of significantly reduced generation due to a combination of weaker sun, cloudier skies, and much shorter days. Designers must prioritize December's output if year-round autonomy is required, accepting that the system will be oversized for the rest of the year.

Task 4: Grid-Connected PV Plant Performance

Objective: Analyze the monthly energy output of the designed PV system and evaluate the impact of temperature.

a) Monthly Energy Output:

- Use PVGIS to estimate the monthly energy output (in kWh) for the grid-connected PV system designed in Task 1.
- Present the results in a table or graph, showing energy output for each month.

✓ Installed Peak PV Power = 104.64 kWp
✓ Estimated System Losses = 14%
✓ Slope of mounted PV grid(opt) = 6^0
✓ Azimuth of PV grid = -90^0

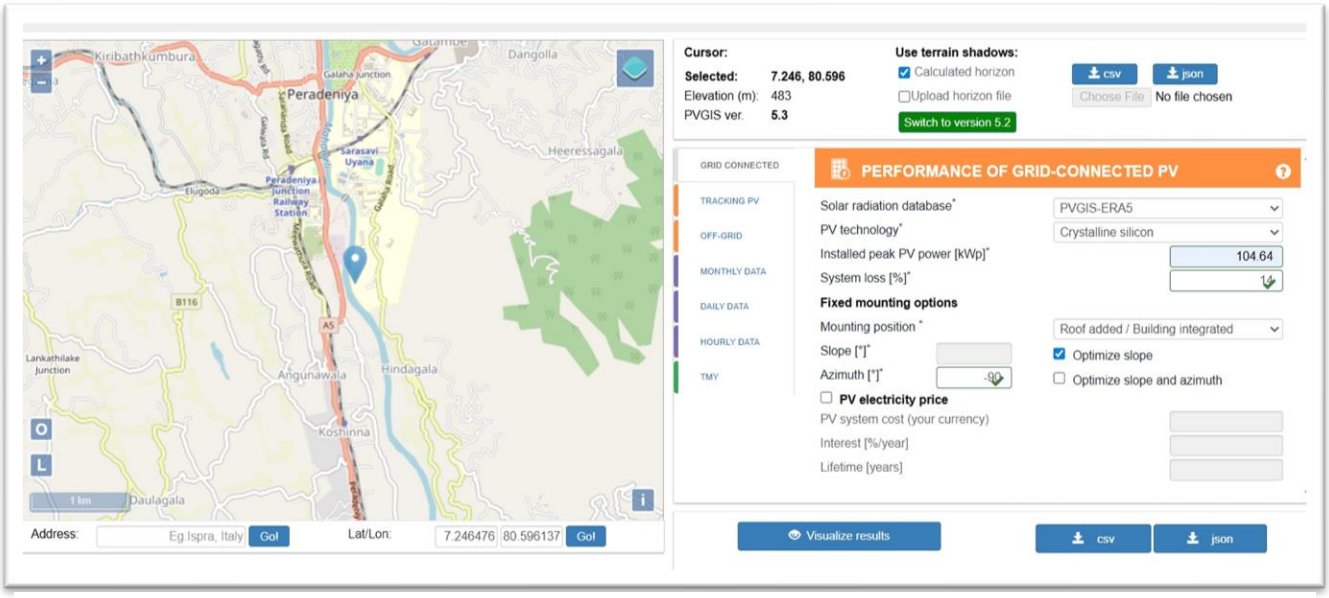


Figure 15 : Selected site with input parameters and coordinates on PVGIS platform for performance evaluation

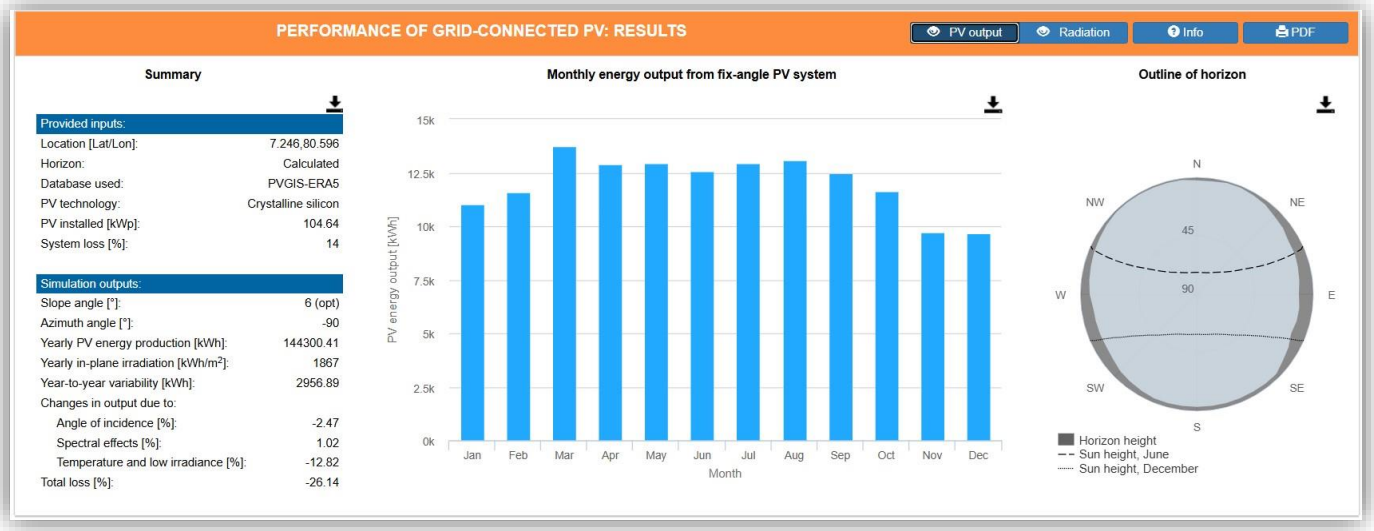


Figure 16 : Simulated Monthly Energy Output of the PV grid using PVGIS tool

Table 5 : Monthly Energy Output of the PV system

Month	Energy Output (kWh)
January	11045.77
February	11566.57
March	13721.12
April	12915.06
May	12935.11
June	12579.95
July	12951.77
August	13083.35
September	12468.68
October	11634.29
November	9739.99
December	9658.75

b) Temperature Impact:

- Obtain monthly average ambient temperatures for the hostel's location using PVGIS or a reliable weather database.
- Compare the monthly PV energy output with the corresponding average temperature.
- Discuss the effect of cell temperature on PV panel efficiency and power output, referencing the temperature coefficient of the selected panels (typically -0.3% to -0.5% per °C above 25°C).

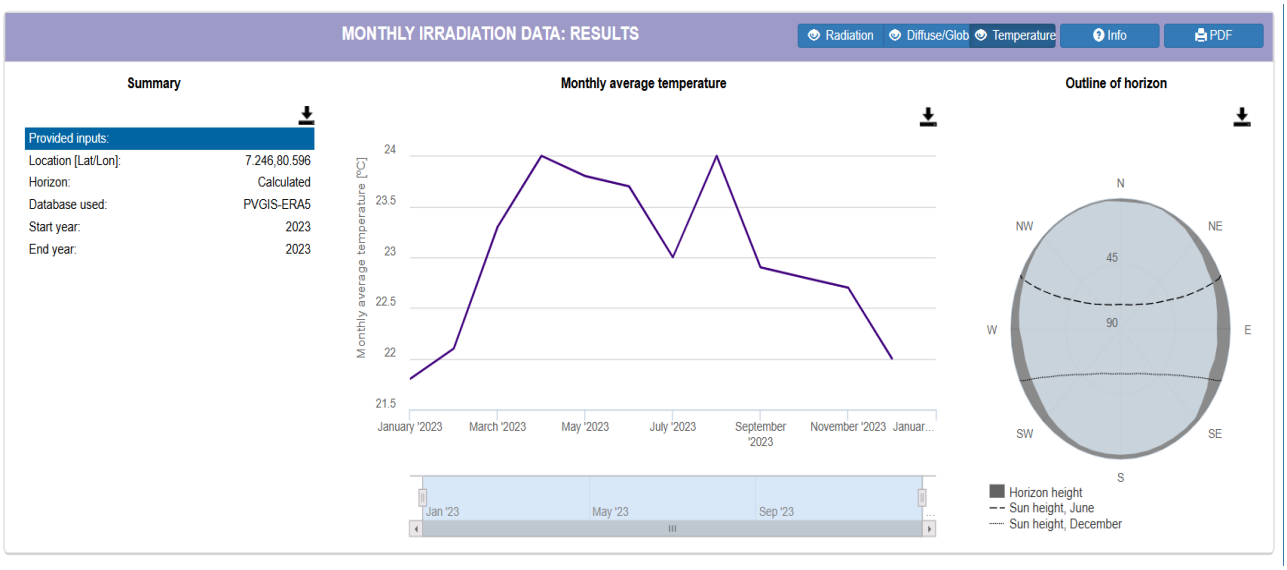


Figure 17 : PVGIS simulated Monthly Irradiation Data and Ambient Temperature Profile of the Location

Table 6 : Monthly Mean Ambient Temperatures at the study site

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Month	Average Ambient Temperature (ºC)
January	21.8
February	22.1
March	23.3
April	24.0
May	23.8
June	23.7
July	23.0
August	24.0
September	22.9
October	22.8
November	22.7
December	22.0

Temperature Impact, Comparison with Energy Output and effect of cell temperature on PV panel efficiency and power output :

The monthly PV energy output follows the seasonal solar radiation pattern, with the highest generation recorded in March (13,721.12 kWh) and consistently strong output from April to August, while the lowest values occur in November and December. When these values are compared with the corresponding average ambient temperatures, it can be seen that the warmer months (≈23–24 °C) coincide with higher energy production due to greater sunlight availability. However, the comparison also highlights that during hot months the expected gain from irradiance is slightly reduced, as the rise in ambient temperature elevates the cell operating temperature well above standard conditions. In contrast, cooler months such as January and December show a lower overall energy output, but this is mainly due to reduced solar resources than temperature limitations.

The effect of elevated cell temperature on PV panel efficiency is explained by the negative temperature coefficient of power, which for most crystalline silicon modules lies between –0.3% and –0.5% per °C above 25 °C. This means that if the cell temperature reaches 45 °C, the panel could lose around 6–8% of its rated output. For the system under study, estimated cell temperatures of 42–44 °C in the hottest months result in efficiency reductions of about 7–7.5%. These losses demonstrate that while **irradiance is the dominant factor** controlling monthly production, temperature imposes a consistent penalty on performance, and its effect should be considered in both system design and yield estimation.

Analysis:

Explain how temperature variations influence the system’s performance and suggest design considerations (e.g., ventilation, panel type) to mitigate efficiency losses.

Temperature changes have a big impact on how well a PV system works. When the ambient temperature goes up, the cell temperature of the modules also rises. It can often reach 15 to 25 °C higher than the surrounding air because of the solar radiation absorbed. Since the power output of most crystalline silicon modules drops by about 0.3 to 0.5% for every °C above the standard 25 °C reference, higher cell temperatures result in clear efficiency losses. For instance, if the cell temperature reaches 45 °C, power output can drop by 7 to 8%. This means that even in months with plenty of sunlight, the actual energy yield is less than the potential, as thermal effects diminish some of the benefits from irradiance.

To lessen these losses, several design factors are key. First, maintain enough airflow behind rooftop panels by leaving a gap between the modules and the roof surface; this helps release heat. Second, choose module types with lower temperature coefficients, like N-type or heterojunction (HJT) panels, which handle heat better. Third, use reflective or light-colored roofing materials to decrease heat buildup under the array. Lastly, check string sizing carefully so that the inverter's maximum power point tracking (MPPT) can work well in both hot and cool conditions. These steps help the system maintain higher efficiency and provide more steady energy output throughout the year.

Task 5: Annual Revenue Calculation

Objective: Estimate the financial benefits of the rooftop solar plant based on energy generation and consumption.

a) Energy Generation and Consumption:

- Use the annual energy output from Task 4 and the daily demand estimates from Task 1 to calculate:
  - Total annual energy generated by the PV system (kWh).
  - Total annual energy consumed by the hostel (kWh).
  - Net energy exported to the grid or imported from the grid Net energy exported to the grid
- Assume net metering tariff (LKR/kWh).

Total annual energy generated by the PV system (kWh).	= 11045.77 + 11566.57+ 13721.12+ 12915.06 + 12935.11+ 12579.95+ 12951.77+ 12951.77+ 12468.68+ 11634.29+ 9739.99+ 9658.75 = 131 365.35
Total annual energy consumed by the hostel (kWh).	= (energy consumption per week day*22*12) + (energy consumption per weekend day *2*4*12) = (360.6*22*12) + (85.2 *2*4*12) = 95198.4 + 8179.2 = 103377.6
Net energy exported to the grid per year (kWh)	= 131 365.35 – 103 377.6 = 27987.75

Total earning = 27987.75 kWh * 15.49 Rs/kWh

= Rs. 433 530.25 /=

NOTE: Here for the Total annual energy consumed by the hostel, it was assumed that there are 22 weekdays and 8 weekends for each month to be form 30 days. And further assumed that each month has 30 days on every months of a year.

b) Revenue Calculation:

- Calculate the annual revenue from:
  - Savings on electricity bills (energy consumed from PV instead of the grid).
  - Income from exporting excess energy to the grid (if applicable).

1.Savings on electricity bills

Total annual energy consumed by the hostel per month,

= 103377.6 kWh / 12

= 8614.8 kWh

Total monthly electricity bill (considering domestic tariff),

= (60*12.75) + (30*18.50) + (30*24) + (60*41) + (8434.8*61) + 5000.00

= Rs. 524 022.8 /=

Annual electricity bill,

= 524 022.8 * 12

= Rs. 6, 288, 273.60 /=

2.Income from exporting excess energy to the grid

Income per year from exporting excess energy,

= 27987.75 kWh * 15.49 Rs/kWh

= Rs. 433530.25 /=

3. The annual revenue

The annual revenue,

= 6 288 273.6 + 433 530.25

= Rs. 6,721,803.85 /=

c) Discussion:

- Discuss factors affecting revenue, such as seasonal variations in generation and demand, or policy incentives.

1. **Seasonal Variations in Solar Generation**
    - Solar power output directly depends on the amount of sunlight received, which varies with season, weather, and geographic location.
    - In summer, longer days and clearer skies usually lead to higher energy generation, boosting revenue. In contrast, winter months with shorter days, cloud cover, or monsoon rains (in tropical regions) reduce production.
    - These fluctuations mean revenue is not constant year-round, and payback periods for investments depend on the annual average generation, not just peak months.
2. **Seasonal Variations in Electricity Demand**
    - Demand for electricity also shifts seasonally. For example, in hot climates, summer air conditioning drives up consumption, which may align well with high solar output. In cooler regions, winter heating demand peaks when solar output is lowest, limiting direct consumption benefits.
    - Where net metering or feed-in tariffs exist, excess summer electricity can be sold back to the grid, stabilizing revenue across the year. Without such schemes, seasonal mismatch can reduce overall financial returns.
3. **Policy Incentives and Regulatory Frameworks**
    - Government subsidies, tax credits, and grants lower the upfront cost of solar installations, improving profitability.
    - Feed-in tariffs (FiTs) and net metering policies allow system owners to earn revenue by selling excess electricity to the grid, significantly impacting overall revenue.
    - Some regions introduce time-of-use (TOU) pricing, where electricity sold during peak demand hours fetches higher rates, increasing revenue potential for solar operators.
    - Policy stability is crucial: sudden changes in subsidies or tariffs can affect long-term revenue forecasts and investor confidence**.**
4. **Market and Economic Factors**
    - Electricity prices fluctuate depending on fuel costs, grid demand, and market conditions. Higher conventional electricity prices make solar energy more competitive, raising revenue.
    - In deregulated electricity markets, solar producers may also sell renewable energy certificates (RECs) or carbon credits, creating additional revenue streams.
5. **System Performance and Degradation**
    - Solar panels degrade over time (typically 0.5–1% per year), which gradually reduces generation and hence revenue.
    - Proper maintenance, cleaning, and use of high-quality components can minimize performance losses, sustaining revenue longer.
6. **Geographical and Environmental Factors**
    - Location affects solar irradiance (sunlight intensity) and thus potential revenue.
    - Areas prone to dust, snow, or shading may experience reduced efficiency, directly lowering financial returns.

Task 6: Financial Analysis of the PV Project
Objective: Evaluate the financial viability of the PV system over its operational life.
Given Data:

- Initial capital cost: LKR 100,000 per kW of installed capacity.
- Annual Operation & Maintenance (O&M) cost: LKR 50,000.
- Inverter replacement cost: LKR 400,000 after 5 years.
- Project lifespan: 15 years.
- Discount rates: 8%.

a) Project Timeline:

- Draw a timeline (e.g., using a Gantt chart or simple diagram) showing:
  - Initial investment (Year 0).
  - Annual O&M costs (Years 1–15).
  - Inverter replacement (Year 5).
  - Annual revenue from energy generation (Years 1–15).

Initial Capital Investment = 100,000 Rs × Installed kW
= 100 000 x 104.64
= Rs. 10, 464, 000.00/=

Table 7 : Cash flow through 15 years (after installation)

Year	Revenue	Maintenance	Replacement	Cumulative Cash Flow
0	-10 464 000.00			-10,464,000.00
1	6,721,803.85	50 000		-3,792,196.15
2	6,721,803.85	50 000		2,879,607.70
3	6,721,803.85	50 000		9,551,411.55
4	6,721,803.85	50 000		16,223,215.40
5	6,721,803.85	50 000	400 000	22,495,019.25
6	6,721,803.85	50 000		29,166,823.10
7	6,721,803.85	50 000		35,838,626.95
8	6,721,803.85	50 000		42,510,430.80
9	6,721,803.85	50 000		49,182,234.65
10	6,721,803.85	50 000	400 000	55,454,038.50
11	6,721,803.85	50 000		62,125,842.35
12	6,721,803.85	50 000		68,797,646.20
13	6,721,803.85	50 000		75,469,450.05
14	6,721,803.85	50 000		82,141,253.90
15	6,721,803.85	50 000	400 000	88,413,057.75

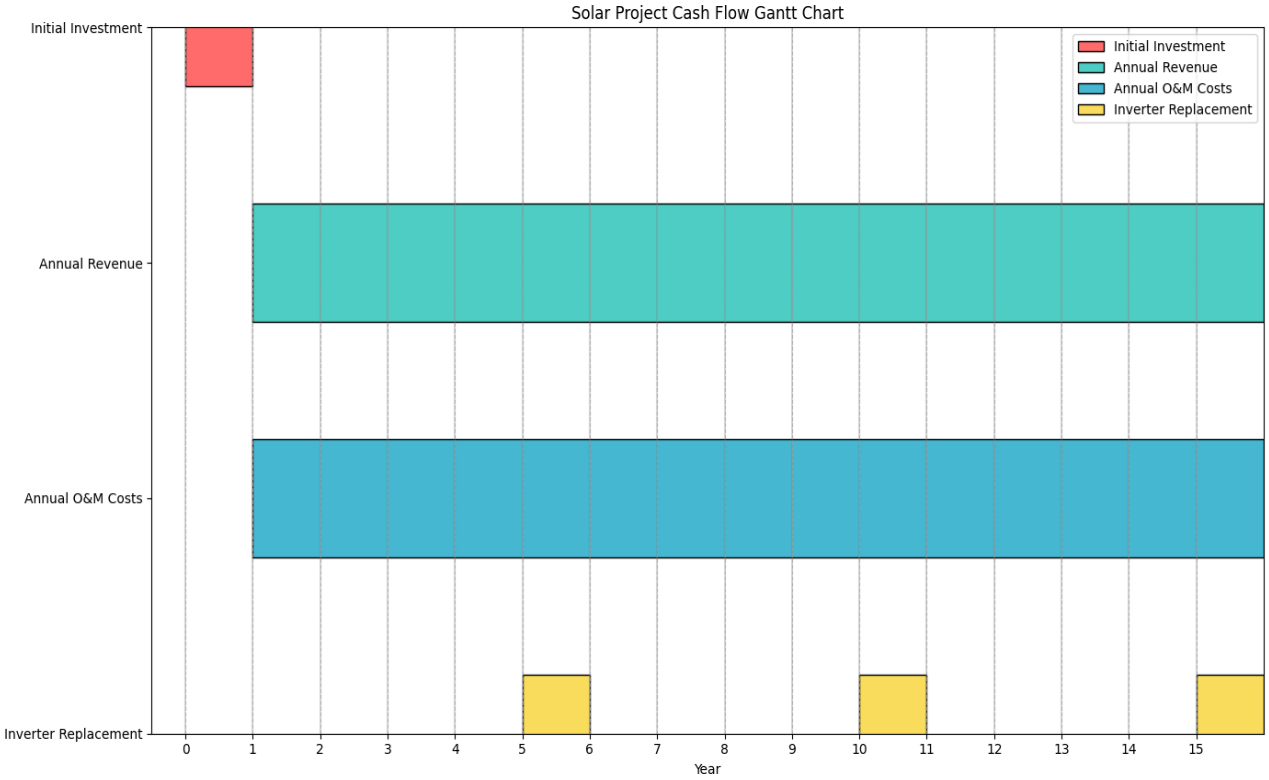


Figure 18 : Grantt Chart of Cash Flow

b) Net Present Value (NPV):

- Calculate the NPV of the project

- Present cash flows in a table, including:
  - Initial investment (negative cash flow).
  - Annual revenue (from Task 5).
  - Annual O&M costs (negative cash flow).
  - Inverter replacement cost (negative cash flow in Year 5).

- Compute NPV and interpret the results.

Table 8 : Present Values through 15 years (after installation)

Year	Cumulative Cash Flow	Annual cash flow	Present Value (PV)
0	-10,464,000.00	-10 464 000.00	-10,464,000.00
1	-3,792,196.15	6,721,803.85	6,177,596.16
2	2,879,607.70	6,721,803.85	5,720,089.04
3	9,551,411.55	6,721,803.85	5,296,378.74
4	16,223,215.40	6,721,803.85	4,904,054.39
5	22,495,019.25	6,721,803.85	454,0791.1
6	29,166,823.10	6,721,803.85	420,4436.2
7	35,838,626.95	6,721,803.85	3,892,996.48
8	42,510,430.80	6,721,803.85	3,604,626.37
9	49,182,234.65	6,721,803.85	3,337,617.01
10	55,454,038.50	6,721,803.85	3,090,386.12
11	62,125,842.35	6,721,803.85	2,861,468.63
12	68,797,646.20	6,721,803.85	2,649,507.99
13	75,469,450.05	6,721,803.85	2,453,248.14
14	82,141,253.90	6,721,803.85	2,271,526.06
15	88,413,057.75	6,721,803.85	2,103,264.87

NPV of the project = 47,544,491.30 LKR

Since the NPV is positive, the project is financially viable under these assumptions.

Group Contribution Statement :-

Student	Name	E - Number	Role	Description	Task	Percentage Contribution To The Project (%)
1	PERERA K.L.D.P. MISS	E/21/295	Financial Analyst	Handles revenue calculations and financial analysis	5	14
2	PETERSON S.D. MR	E/21/303	Performance Analyst	Analyses energy output and temperature effects	4	14
3	PRABHASHANA D.M.M. MR	E/21/311	Solar Resource Analyst	Manages data collection and visualization using PVGIS	2	14
4	RATHNAYAKA P.G.I.N.B. MR	E/21/327	Project Engineer, Design Engineer	Leads the solar plant design, including panel selection, system sizing and electrical layout	1 (c,d)	14
5	SANGANI Q.A.L.R.M. MR	E/21/355	Solar Resource Analyst	Manages data collection and visualization using PVGIS	3	15
6	SELVAKUMAR K. MISS	E/21/367	Financial Analyst	Handles revenue calculations and financial analysis	6	15
7	VIDANAPATHIRANA C.J. MR	E/21/419	Design Engineer	Leads the solar plant design, including panel selection, system sizing and electrical layout	1 (a,b)	14
Total						100%